Chapter 1: Introduction to ECON 1630

Jonathan Roth

Mathematical Econometrics I
Brown University

Outline

1. Course Preliminaries

2. What is Econometrics?

3. Why is Econometrics Challenging?

4. Course Roadmap

Introducing Ourselves

Welcome to ECON 1630! I'm looking forward to teaching you all

I'm Professor Jonathan Roth

- Group OH on Zoom (link): Mon 215-245
- Individual OH: Typically Tue 3-4; sign up on my website: here, 8 Fones Alley 014 (or Zoom by request)

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Our Grad TAs are Sharad Hotha and Max Grozovsky
Our undergrad TAs are Rohan Pajwani and Marta Jurzyk (star students from previous semesters!)

- All TAs will hold OHs
- Grad TAs will hold weekly sessions
  - Review material and teach coding in Stata
- Times/locations for TA OHs and sections will be announced shortly on Canvas

Canvas and EdDiscussion

Course materials and communications will be posted on Canvas:
`https://canvas.brown.edu/courses/1104214`

The Canvas page has an EdDiscussion board, which is a great place to ask (and answer) questions. The TAs will monitor the Qs.

Meeting times:

- Lectures: Mon/Wed 830-950 (S02) or 3-420pm (S01).
  Recordings will be posted online after class.
- TA Sessions: Time/location TBD
- **Attendance** is not required but is **highly encouraged**

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- Some familiarity with reading/writing proofs and code

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Software:

- Default is Stata for statistical analyses (to be covered in TA sessions)
  - You're welcome to use R instead; TAs are familiar with R
- LaTeX for typing up problem sets (optional, for extra credit)
- Ask us for help if you're having any problems accessing software

Assessments:

- 6 problem sets due approximately every 2 weeks, submitted via Gradescope
- 1 midterm exam (November 4, in class)
- 1 final exam

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Grading:

- 30% problem sets, 35% on each exam.
- I'll drop your lowest PSet grade. Use your drop wisely!
- Psets are due at 4PM on Fridays; late submissions won't be graded. Collaboration is OK (please list collaborators)
- The exams will be in-class, closed-book, "cheat-sheet" allowed
- 5 points extra credit if you use LaTeX for assignments. Please attach your code + output as a single PDF regardless

AI Policy

My view is that AI is a useful tool and we shouldn't ignore it. But need to use it in a way that enhances rather than impedes learning.

- **Exams (70%):** closed-book; *no AI allowed*.
- **Problem sets:** AI OK if it *helps you learn*, not if it replaces learning/effort.
  - **Good:** debugging R/Stata errors.
  - **Bad:** writing free-form answers.
- I can't police what you do on the psets, but if you don't put in effort you probably won't do well on the exams
- Include brief **AI use disclosure** on each pset.

Course materials:

- Main material: Lectures and lecture slides, which will be posted on Canvas
- Optional text: Stock & Watson – Intro to Econometrics (4th ed)

Any questions on logistics?

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2. What is Econometrics?
3. Parameters, Estimands, and Estimators
4. Course Roadmap

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In this course, we will focus mainly on descriptive and causal questions, with an emphasis on causal questions

Why is answering these questions hard?

- For descriptive Qs: we only observe data for a **sample** of individuals, not for the full **population**
  - Example: we want to know how the distribution of income in the US has changed. But we only observe income for a survey of workers

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  Our sample is **randomly** selected from the population
  - E.g., the workers in our survey were drawn out of hat with names of all possible workers
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- Worst case scenario: our sample is *not representative* of the population that we care about
  - E.g., workers with certain characteristics were more likely to respond to the survey



Chicago Daily Tribune

THE WORLD'S GREATEST NEWSPAPER

HOME

DEWEY DEFEATS TRUMAN

G.O.P. Sweep Indicated in State; Boyle Leads in City

REPUBLICAN TICKET AHEAD OF 1944 VOTE

Town Balloting Gives Trend

Tops Coghlan in Hot Race for Attorney

RECORD CITY VOTE SEEN IN LATE TALLIES

Suburban Ballot Near 375,000

BULLETINS ON ELECTIONS

Early Count Gives G.O.P. Senate Edge

PUTS G.O.P. BACK IN THE WHITE HOUSE

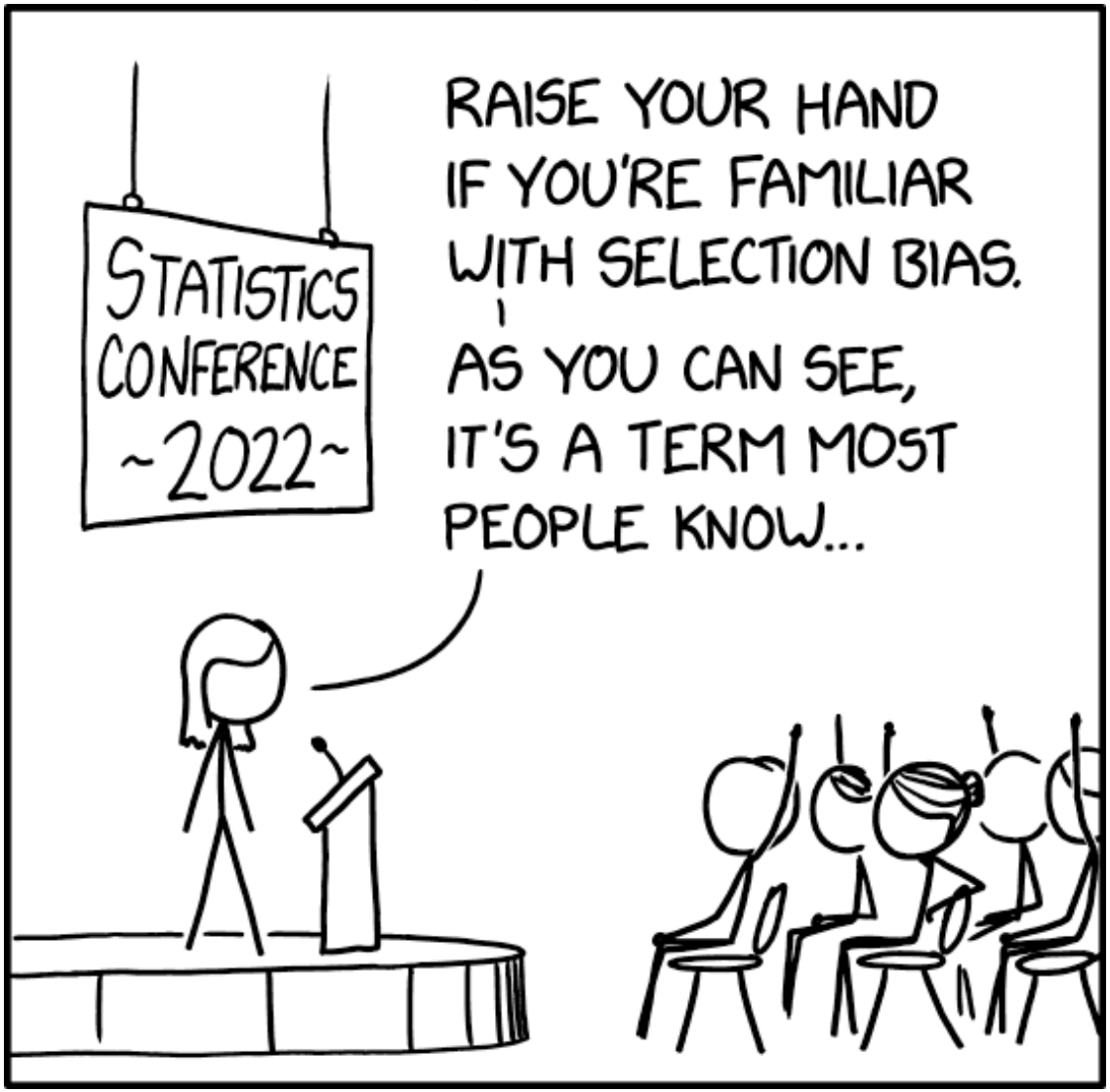
Sizable Electoral Margin Seen

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- But their survey was conducted by phone. In 1948, only rich people had phones: sample $\neq$ population $\rightarrow$ misleading results!

Selection bias referes to settings like Dewey-Truman where the sample is not drawn randomly from the population of interest



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- Counterfactual Qs can't ever be answered with data alone. Need additional assumptions to learn about them!

Splitting up the problem

- When thinking about causal Qs, it's often easier to split the problem into the *counterfactual* and *descriptive* parts

- **Identification:** assume first that we have access to the observable data for the entire population. What can we learn about the parameters we care about (causal effects)?
  - Need to describe how to learn *counterfactual* outcomes from the data we actually observe

- **Statistics**: what can we learn about the full population that we care about from the finite sample that we have?
  - This is the *descriptive* part.
  - Need to understand the process by which our data is generated from the full population

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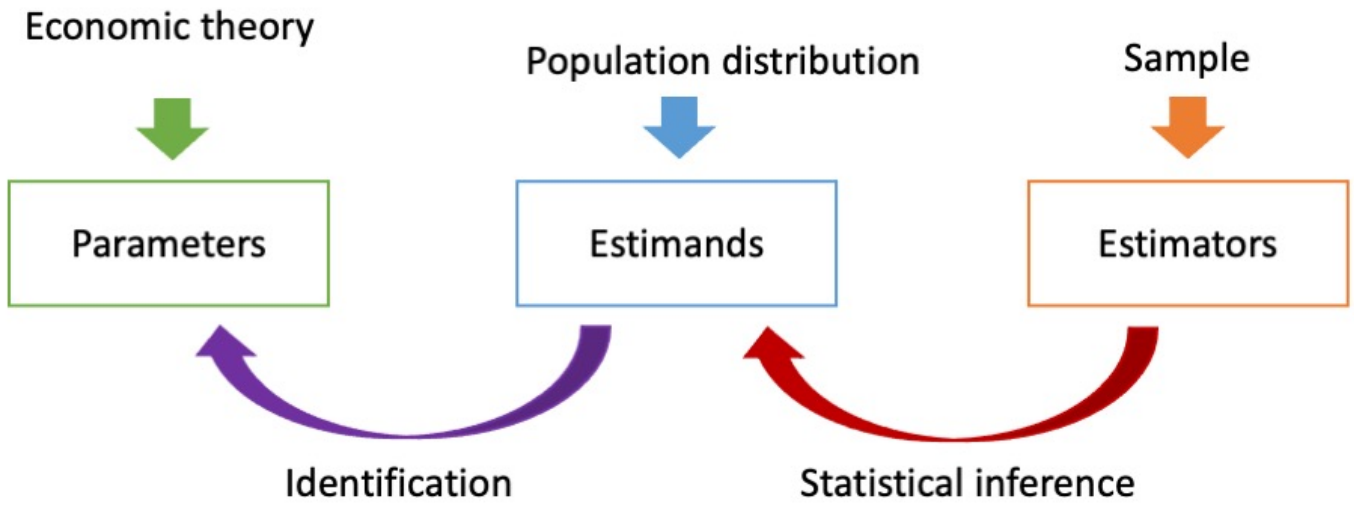
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- The process of learning about the *parameter* from the *estimand* is called **identification**.


Economic theory
Population distribution
Sample
Parameters
Estimands
Estimators
Identification
Statistical inference

Let's add some math...

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- We can write the observed outcome as $Y_i = D_i Y_i(1) + (1 - D_i) Y_i(0)$

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- Example target parameter:
  - Causal effect of Brown for Brown students:

$$\underbrace{E[Y_i(1)|D_i=1]}_{\text{Earnings at Brown for Brown students in pop}} - \underbrace{E[Y_i(0)|D_i=1]}_{\text{Earnings at URI for Brown students in pop}}.$$

Why is causal identification hard?

- Thought experiment: suppose we had data on earnings for *every* Brown and URI graduate

- We can learn from the data:

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- The data doesn't tell us $\underbrace{E[Y_i(0)|D_i = 1]}_{\text{Earnings at URI for Brown Students}}$. Why not?
  - Because we never see Brown students going to URI!

- One idea to solve this problem would be to assume that:

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- Why might this give us the wrong answer?
- Because Brown students may be different from URI students in other ways that would affect their earnings (regardless of where they went to college)
  - Academic ability, family background, career goals, etc.
- These differences are referred to as *omitted variables* or *confounding factors*

What about experiments?

- The gold standard for learning about causal effects is a randomized controlled trial (RCT), aka experiment
- Suppose that the Brown and URI administration randomized who got into which college (assume these are the only 2 colleges for simplicity)
- Since college is randomly assigned, the only thing that differs between Brown and URI students is the college they went to
- Hence,

$$\underbrace{E[Y_i(0)|D_i = 1]}_{\text{Earnings at URI for Brown Students}} = \underbrace{E[Y_i(0)|D_i = 0]}_{\text{Earnings at URI for URI Students}}$$

since we've eliminated any confounding factors

But running experiments is often hard/impossible

- Unfortunately, Brown/URI have not let us randomize who gets into which college
  - At least not yet! If you could convince them to do this, it'd make for a cool senior thesis!
- Likewise, it is difficult to convince states to randomize their minimum wages, or other policies
- In some cases, randomization is not just difficult but would be immoral
  - "What is the causal effect of spousal death on labor supply?"

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  - "What is the causal effect of spousal death on labor supply?"
- In this course, we'll discuss tools economists try to use when running experiments is not possible.

Course Roadmap – Where we're going

- **Part I ($\sim$ 7 lectures): Review of probability/statistics**. This will give us a mathematical language to talk about:
  1. *Statistical estimation/inference:* how does the sample we observe relate to the population of interest
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- **Part III ($\sim$ 7 lectures:) Other "quasi-experimental" strategies**: We'll discuss other strategies for "mimicking" an experiment when it's not available, including instrumental variables (IV) and regression discontinuity (RD)